

MSc Data Science Guidance on The Extended Research Project

Year 2025/2026

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1. Introduction

Students on the MSc in Data Science are required to conduct an extended research project on a topic approved by their pathway lead. The extended research project contributes a third of a programme's assessment (60 credits out of 180). Extended research project reports are normally submitted at the end of the MSc year (in early September).

During the autumn semester, each pathway will provide their students with a list of possible projects. You are also free to develop your own project, subject to approval which will consider:

1. whether there is relevant supervisory expertise within their pathway
2. the feasibility of the topic within the timescale of the programme
3. the extent to which the topic supplements and extends knowledge gained from following the taught component of the MSc.

However, the projects that are provided by your pathway are broadly scoped and allow plenty of capacity for students to bring their own ideas so in most cases developing a project outside of this will be unnecessary.

These guidelines have been produced to help you with the extended research project process. The document sets out the initial objectives and requirements of the project, advises on report preparation, outlines formatting and submission arrangements for the report and details the assessment criteria. You should use it as an aide-mémoire alongside your Programme Handbook and the advice of your supervisor.

1.1 Scope of an extended research project

The overall aims of the extended research projects are to:

- provide students with an opportunity to plan, manage and conduct a programme of research on a topic related to their programme of studies.
- further students' knowledge of a relevant body of data science methods and knowledge, and to develop powers of critical reasoning.
- enable students to seek new research findings which add to the existing body of knowledge on the project's subject.
- develop students' knowledge of, and competence in, an appropriate range of research methods, including the development of research questions and hypotheses, an appreciation of the research methodology and analytical techniques to be utilised, the undertaking of a specific research study, the synthesis and evaluation of findings, and a clear statement of conclusions and recommendations.
- Develop students' understanding of what is needed to ensure a project is reproducible.

- develop students' writing, presentation and referencing skills.
- develop students' experience of developing and managing a specific programme of work through to final submission.

An extended research project requires a total input of 600 hours of student time, comprising 100 hours during the second semester and 500 hours during the period June-September.

1.2 The assessment elements

Assessment task <i>(Please include opportunities for formative feedback)</i>	Length required	Weighting within unit (if relevant)
Ethics Form	1-2 pages	-
A project plan.	1000 words	15%
A report on the project, its methods, and outcomes. This could either be in the form of a technical report or an academic paper.	<7500 words	60%
A presentation – a three - five-minute recorded presentation outlining the aims of the project, the approach and the results – either in the form of a video poster or an annotated slide deck.	<5 minutes	10%
A set of supporting materials which support reproducibility; a technical appendix and/or a code repository.	varied	15%

The rubric for the report is contained in section 4. The assessment criteria for other tasks will be provided at the first ERP training session.

1.3 Submission arrangements

The submission of the final research project outputs will mark the completion of your degree, and we need the time following the deadline to mark, second mark, review and process your grades in time for the Exam Board in the Autumn and allow time for graduation in December.

Programme Ethics form (by email to Mark Elliot) by 2pm on **24th March 2026**.

Project Plan via Canvas by 2pm on **31st March 2026**.

Report via Turnitin via Canvas by 2pm on **7th September 2026**.

Additional materials via canvas on Canvas by 2pm on **7th September 2026**.

Video submitted (mechanism TBC) by 2pm on **14th September 2026**.

Submission arrangements for students with summer resits

Students who have failed any postgraduate course unit and who are completing summer resits for course work or examinations will, by default, have the deadline for report submission and video deferred to 2nd December 2025. This enables the student to be sure they have successfully completed all their other credits before focusing on their project.

1.4 Penalties for Late Submission

In accordance with Faculty policy, any student who submits their project report after the submission deadline will receive a mark of zero for the report, unless they are subsequently able to prove mitigating circumstances.

Regulations about your assessment can be found in the SoSS Handbook on the Data Science Community space on blackboard. It is your responsibility to ensure that you are fully aware of these regulations. If you are unclear regarding any aspect of the regulations, then seek advice from your Pathway lead.

Where relevant, students should alert their supervisor to any extenuating circumstances well in advance of the submission deadline.

As with coursework assessment, the Mitigating Circumstances mechanism also applies to extended research projects. Any student who considers that their research project report (or other outputs) may be delayed due to 'unforeseen' and 'unpreventable' circumstances should make their case using the online form (via the link below), with supporting evidence.

<https://www.socialsciences.manchester.ac.uk/student-intranet/postgraduate/postgraduate-taught/ma-msc-diploma-forms-and-guidance/>

All work to be considered under mitigating circumstances should be submitted as soon as is practicable but note that any work submitted after the end of September 2026 may be too late to be marked and considered in time for the December graduation in 2026.

Students who do not submit a report will normally be awarded a Postgraduate Diploma provided they have met all the other requirements of their programme. Students who fail the extended research project will be granted the opportunity to resubmit provided that their mark is 30 or above.

1.5 Assessment arrangements

The project plan, additional materials and video will be assessed by the pathway lead and moderated by the programme director.

The report will be assessed by a minimum of two internal examiners. A sample of extended research projects will be sent to the external examiner for the programme, who validates standards.

Final marks are confirmed by the Board of Examiners, which meets in mid-November. A full explanation of the assessment criteria for the report is set out in **Section 4**.

1.6 Further Reading

There is an extensive range of reading material associated with preparation of research reports. Some useful texts are:

- Allan, G. and Skinner C. (eds.) (1991) *Handbook for Research Students in the Social Sciences*, The Falmer Press, London.
- Berry, R. (1994) *The Research project: how to write it*, Routledge, London.
- Bouma, G. and Ling, R. (2005) *The Research Process*, OUP, Oxford.
- Burton, H.M. (2021) *Your First Research Paper: Learn how to start, structure, write and publish a perfect research paper to get the top mark*.
- Denscombe, M. (2001) *The Good Research Guide*, Open University Press, Buckingham.
- Gastel, B. and Day, R.A., (2022). *How to write and publish a scientific paper*. Bloomsbury Publishing USA.
- Higham, N. J. (1998). *Handbook of writing for the mathematical sciences* (Vol. 63). Siam.
- Hofmann, A. H. (2023) *Scientific Writing and Communication: Papers, Proposals, and Presentations*, Oxford University Press, Oxford.
- Joyner, R. L., Rouse, W. A., & Glatthorn, A. A. (2018). *Writing the winning thesis or dissertation: A step-by-step guide*. Corwin Press.
- Turabian, K. L. (2018). *A manual for writers of research papers, theses, and dissertations: Chicago style for students and researchers*. University of Chicago Press.
- Zaumanis, M. (2021) *Write an impactful research paper: A scientific writing technique that will shape your academic career*. Peer Recognised

2. Research project planning

This section offers some pointers on drafting your report. It contains important information relating to the initial process of preparation and (in **Section 2.6**) the structure of the final report.

While many of the points listed below apply to virtually all pieces of research, a few will be of little relevance to your chosen area of study. This emphasises the importance of consulting your supervisor as a means of gleaning advice related specifically to your project. Nonetheless, many of the suggestions listed offer a general insight into the rules and conventions of research. **It is essential, therefore, that you read (and re-read) these carefully.**

The timetable for extended research project topic selection, supervisory meetings and submission is provided in **Appendix 1**.

2.1 Keeping on top of the research project process

Planning ahead

When you are planning your schedule, please do remember that your own time and effort is only one of the determining factors. You need to consider your supervisor's time as well as your own when estimating overall time needed. Primary, data collection, if that is part of your project, always takes longer than one expects, as does, for example, the writing of a literature review and even the completion of your reference list, so be on the safe side and double your first estimate.

Organising and backing up your work

Organising your work is of key importance. You should build up separate reference files of material as the study develops including notes of all documents read, and photocopies of particularly useful papers or diagrams, and notes of all meetings and discussions. Ensure that all references are complete. Allow time for the development of ideas and arguments through revision and redrafting and full discussions of each section of study. Preparing summaries of each section as part of the drafts can help you achieve this.

Remember to make back-up copies of any work held on hard or portable disks. This can be done using your university storage space or a free web-based cloud service like Dropbox or Google Drive. If you have a PC or laptop, backup copies using some of these alternatives. Note that any such 'disasters' of lost work will not justify late submission. Make sure you leave plenty of time to proofread work, format the layout, chase-up any remaining references and print the final copies; these invariably take much longer than you expect.

The writing process

A golden rule is to write down as much as you can from as early on as you can. There are several good reasons for this advice.

Quite often, it is difficult to spot the flaws in your reasoning until it is set out on paper, so you are well advised to write up bits and pieces in draft as early as possible. An early session with your supervisor should be followed by a period where you develop – and get down on paper – ideas that have been discussed.

Another reason for writing early on is that your supervisor will want to see your reasoning is committed to paper; supervisions based on loose discussion can be a waste of time. Some supervisors say that the most frequent advice they give students is to “go away and write down everything you have just said”.

A third reason for writing things down is reduce the burden on the main piece of equipment that you will be using – your brain! If you are carrying ideas, tasks, and actions around in your head then you will be constantly cycling through them. Externalising them will reduce the stress that you will undoubtedly experience and free up mental bandwidth to focus on the task that you are doing right now.

For some people, getting thoughts and methods down on paper presents one of the biggest stumbling blocks. If you feel this way, take comfort that this is a common feeling. However, this must be overcome early in the process.

One of the reasons why you might be reluctant to do this is lack of confidence. Many people who have not produced an extended piece of writing before are diffident about committing their critical thoughts and methodology descriptions to paper. A more positive reason for getting down to it early is that you will be pleasantly surprised at how many words it takes to put down your ideas. The daunting task writing does not seem nearly so unattainable when you realise how many thousands you have already written on the literature review and a description of methodology. Indeed, it is not uncommon for student to find themselves overlimit and needing to cut down towards the end of the process.

2.2 Working with your Supervisor

The student's role

The initiative for requesting supervision sessions lies entirely with you, the student. Agree methods of getting in contact with your own supervisor: email is usually the best way, and if you wish to see your supervisor, you should make an appointment. You must ensure that your project supervisors are kept fully informed on progress and difficulties and are ‘interviewed’ with prepared questions at supervisory meetings. You should submit chapter drafts in sufficient time in advance of meetings to allow for comment and discussion before proceeding to the next stage.

Your supervisor will almost certainly be supervising other students and therefore, you should not assume that s/he can immediately recall the last discussion you had together about yours. Supervisions always need to be pre-arranged. You should never expect on-the-spot supervisions. There will also be periods when your supervisor is not available, either because s/he is heavily committed with other duties such as examining, or on leave for work or holiday. Do make sure that you and your supervisor are aware of each other's periods of absence.

Supervisors cannot judge how your work is going if you just hand over a sub-section that you have written up. You need to create at an early stage and then regularly update a draft of your report's table of contents so that it is quite clear what follows on from what. You may also need to add notes to show if and how the section in question is incomplete.

Note, supervisors are not to be expected to proofread or to correct spelling/grammar. Students are advised to buy-in or otherwise arrange such services if needed. Even if your supervisor reads your entire report and you make the suggested changes, this is no guarantee that it is of a pass standard; after all, the work is yours and its quality is dependent on your output.

To a large extent, then, your project must be a self-managed process. Your role is to organise the research programme, taking advice from your supervisor and taking the initiative in raising problems/difficulties. The supervisor's role is to give advice and help about the nature and standard of the work and direct you to useful literature and appropriate methodology. But remember, the ultimate responsibility remains yours. Do not expect your supervisor to read multiple re-drafts of every piece of your work, and do not embarrass your supervisor in the latter stages by asking whether you will be successful. When submitted, the report is referred to internal and, in some instances, external examiners who will make an independent judgment of your work in its entirety.

The role of the supervisor is to:

- establish at an early stage their responsibilities in relation to the student's written work, including the nature of guidance and comments to be offered as work proceeds.
- provide advice and guidance to help improve the quality of the work.
- be available, where possible, for meetings which should be agreed in advance.
- give guidance concerning the nature of the research, providing constructive feedback on the standard of work expected and helping to plan the programme of research involved.
- agree completion dates for successive stages of the work, requesting draft chapters as appropriate and returning written material with constructive criticism on the broad shape and structure of the work (but not on its detailed content).

Supervisory structure

Over the course of the second semester and through the summer period prior to submission, supervision will comprise a series of individual meetings.

After you have been allocated to a supervisor/project, your first task is to ensure that you are aware when the first supervision session will be. Your supervisor will contact you about this, but please email them if you are in any doubt. Early supervision sessions focus on assisting you in the framing of your research, and in particular the development of your specific research questions. This will lead to the production of your project plan. Your supervisor will advise you on the specific content of this.

Note that the norm on this programme is five supervision sessions prior to the end of June; some of the sessions may be joint with other students who are working on related projects.

There is an onus on you to use this time effectively, and you need to come to each meeting prepared. Try at all costs to avoid rambling supervisions with vague discussions; both of you need to use time economically. Remember that every time you give your supervisor something to read, s/he will need a few days to read it. Supervisions should almost always be based on something written by you, so that the supervisor can give you feedback on your ideas, proposed methods, or analyses. The initiative should be taken by you rather than your supervisor.

During the summer period, extra supervisory meetings may be arranged after consulting your supervisor for availability. During these months you should be either running analyses or writing-up the project.

2.3 Ethical and risk assessments

It is best to consider risk and ethical review as soon as is practicable. You will have a session on this in the ERP training in semester 2.

The first stage is to complete the [ethics decision tool](#) and then incorporate this into the programme ethics review form that you can find on the extended research project BB page. You will be given further guidance on completing this during the ethics session of the research training classes.

Please note that completing and gaining approval from your supervisor does not provide ethical approval for your project. The project will need to be signed off by the programme director as well. The purpose of the form is to help you to identify whether your project raises ethical issues and therefore requires further ethical review. Most projects on this programme should not involve more than minimal ethical risk and therefore should be able to be able to be approved at the programme level.

If you do not need further ethical review, you can proceed with your project. If the process has determined that you require ethical approval then you need to log on to the online [Ethical Review Manager \(ERM\)](#) to create a new ethics application and have it

reviewed and approved so you can start your research. You will be advised on this by the programme director. Without ethical approval, you cannot start your research. You will find full guidance documents and training videos on how to complete the application on the ERM site.

- [Student Ethical Guidance](#)
- [On-line System for Ethical Review \(ERM\)](#)

Risk assessment

The University has a duty to protect your safety. For most data science students, the project will not pose any safety issues. However, if your project requires you to work off campus (other than at home) for example at a company's offices, then you may also need to complete a risk assessment.

As SoSS is the home school for the programme you would use their assessment procedures. The School's generic risk assessment documents are listed below. If your project does not fall within these generic risk assessments then you will have to complete a full risk assessment form, with guidance from your supervisor, outlining the specific risks you anticipate and the measures you plan to take to address them. Submit this risk assessment with your application as an attachment to your application in the ERM. If one of the generic templates fits your project in most, but not all, respects, you can use the full risk assessment form to state which generic assessment mainly applies and to then specify the risks associated with your project that fall outside the generic assessment, and the measures you plan to take to address them.

- [SoSS Generic Risk Assessment A](#)
- [SoSS Generic Risk Assessment B](#)
- [SoSS Generic Risk Assessment C](#)
- [Full Risk Assessment](#)

University documentation

- [Health & Safety in Fieldwork](#)
- [Guidance on Lone Working](#) – If you are going to work alone, you should think about associated risks to the researcher and sensible safeguards for those risks
- [Research Data Management Policy](#) – Ethics approval involves issues of data management (what you will do with your data to secure them and protect them)

Further information on good research conduct, misconduct and policies and guidelines can be located on the Research governance, ethics and integrity webpages: [Research governance, ethics and integrity](#).

Further information related to ethics	
Source of information / act	URL
Disclosure and Barring Service (DBS)	www.gov.uk/disclosure-barring-service-check

Central Office for Research Ethics Committee – COREC (NHS)	www.corec.org.uk
The Human Rights Act (1988)	http://www.legislation.gov.uk/ukpga/1998/42/contents
Data Protection Act (2018)	http://www.legislation.gov.uk/ukpga/2018/12/contents/enacted
UK Copyright Act (1988)	http://www.legislation.gov.uk/ukpga/1988/48/contents
Race Relations Act (1976)	http://www.legislation.gov.uk/ukpga/1976/74/contents
Race relations (Amendment) Act 2000	http://www.legislation.gov.uk/ukpga/2000/34/contents
Disability Discrimination Act (1995)	http://www.legislation.gov.uk/ukpga/1995/50/contents
Freedom of Information Act (2000)	http://www.legislation.gov.uk/ukpga/2000/36/contents https://ico.org.uk/
Communications Act (2003)	http://www.legislation.gov.uk/ukpga/2003/21/contents
University of Manchester • Code of Practice for Dealing with allegations of Misconduct in Research • Disability Discrimination Act Policy • Equality & Diversity Policy • Freedom of Information Act Policy • Health & Safety Policy • Harassment, Discrimination & Bullying Policy • Intellectual Property Policy (guidance on) Plagiarism and other forms of academic malpractice	www.campus.manchester.ac.uk/studentnet/policies/
University's data protection policy	http://www.dataprotection.manchester.ac.uk/

3. Submission format for the report

3.1 General information

All reports must be written in English; quotations, however, may be given in the language in which they were written.

3.2 Presentation

Marks will be awarded for the degree of professionalism in the style and layout of the report. The overall structure must be clearly presented (e.g., with an organised hierarchy of fonts and typefaces for chapter and section headings), with logical layout of sections and paragraphs, and with text and graphics integrated in an overall 'house' style. A worthwhile activity is to review (from a presentational viewpoint) articles that you are citing or others from the field as this will give you some idea what a well-presented article looks like.

3.3 Style and Language

The aim should be to use simple prose, but with variety in the construction of sentences and an expansive approach to the vocabulary employed. Sentences are best kept short (maximum of around three lines), but their length should be varied to avoid monotony. Paragraphs should be of reasonable length (normally 3-6 sentences in length) allowing the reader time to digest one idea or theme before introducing another. Conventions vary by discipline about whether one should write in the first or third person and whether to use the active or passive voice. A good rule of thumb is to aim for a tone and style like that used in the main references you will be citing. Consult your supervisor if unsure.

It is important to adopt a mode of writing which keeps the reader interested, and this can be achieved more easily if the active voice is used (e.g., 'examination of the site revealed...'). Try to adopt this mode of writing right from the start since altering a whole draft can be time consuming. It is also important to use language which is neutral when matters of race and gender are involved. Terminology is often used as verbal shorthand to convey complex ideas and terms employed must be used accurately in the sense by which they are understood among those familiar with the subject.

The most frequently used abbreviations include: *i.e.* (that is), *e.g.* (for example), *etc.* (*et cetera*, other things of the same class), *viz.* (namely), *cf.* (compare with), *no.* (number), *ibid.* (the same place), *idem.* (the same), *sic* (*sic passim*, thus, typically used to denote an error in a quote), *et al* (and others) – be sparing in their use in the main text. Where title abbreviations are used it is common practice to use the full term followed in brackets by the abbreviation on the first occasion of use in the text, and thereafter use the abbreviation only: e.g., Local Enterprise Partnerships (LEPs), Markov Chain Monte Carlo (MCMC). The punctuation between the initial letters of well-known organisations should be omitted in the text (e.g., EU, UN). Do not abbreviate units of measurement unless they are preceded by an exact number e.g., 17ft; do not add an s to the plural of

an abbreviation e.g., 40cm, 18lb. Where you use a local currency, always provide an international currency equivalent – typically US\$ but GB£ or euros are acceptable.

3.4 Word limit

The maximum word count for report is 7,500 words. This should not be exceeded.

The **word count includes**: all text, from the title of chapter one to the end of the last chapter, including the body of text, chapter and section titles, table and figure captions, chapter footnotes and endnotes, quotations, tables, and figures.

The **word count does not include**: the reports title, table of contents pages, declaration of originality, lists of tables and figures, acknowledgements, abstract, the reference list and appendices.

You **must** include a word count on the bottom of your contents page.

What are the penalties for exceeding the word count?

- Markers can consider minor transgressions of up to 10% within the existing marking criteria, which means that you can lose marks for not being concise and exceeding the word limit.
- If you exceed the word count by between 10 and 50%, your final report mark will be capped at 50%. You will still be able to pass the extended research project element, but your mark may be lower than you would have otherwise achieved
- Work exceeding the word count by more than 50% will be viewed as not having met the requirements of the assessment. The work will not be marked and a mark of zero will be recorded.

The implications of this are that if you exceed the word count by more than 50% you will be awarded a zero mark, with no opportunity for a resit and will complete the programme with the award of Postgraduate Diploma.

3.5 Graphical material

Maps, statistical tables, figures, diagrams, graphs, and photographs often provide a useful means of summarising complex information. They can also add to the work in a presentational sense. However, you should take care to use these in a manner appropriate to the topic. Any graphical figures must be referred to in the main body of text and properly labelled. All illustrations must have a designation, number and title situated immediately above or below, usually with the prefix of Figure or Table in capital letters. Illustrations may be lifted from other sources, provided that the original authors/creators are properly acknowledged.

Statistical tables or graphs should normally be no greater than a single A4 size page. All rows and columns should have unambiguous headings and use ruled lines sparingly. Graphs should only have as many grid lines as are needed for comprehension of trends and relationships (normally 4 or 5 being the maximum), and with legends and descriptive

notes normally standing clear of the grid lines. Think carefully about the labels, legends and titles in tables and charts. A good rule of thumb is that any diagram or figure should be understandable independently of the text.

Images, maps, figures, and graphs must be inserted in their final location in the text. A responsible use of these elements is advisable. The use of words in graphical elements (such as infographic or maps) will be checked by the markers and the student may be requested to present a count of these words. Tables, containing both numerical or text information, must not be inserted as images and will be included in the word count.

3.6 Proof reading

The final draft of your report should be read, where possible, by another person to eliminate errors of syntax and grammar, and any typographical errors which are not picked up by standard word processor spell-checkers. Proof reading is time consuming, but extremely important. You should take care to avoid long, rambling sentences, pretentious and jargon-ridden prose, and bad grammar. Learn how to use the apostrophe in the correct manner. Avoid vague, vogueish jargon like ‘sustainability’, ‘partnership’ or ‘stakeholder’ unless you are sure of the precise meaning of these. Note that it is not your supervisor’s role to proof-read final drafts.

3.7 Plagiarism

The University’s regulations covering plagiarism (copying work from others without reference to the source) is fully set out in your programme handbook. It is regarded as a serious malpractice and may lead to severe penalties for the work submitted as well as being recorded on the student’s record. The most common way in which plagiarism occurs is by the verbatim reproduction of another author’s work without acknowledgement, or the ‘lifting’ of a concept from a specific source without attribution. Reports which contain long quotations or illustrations which are not the writer’s own work cannot be published without the express consent of the copyright holder, although extended research project reports are not generally considered to be ‘publication’. Full and proper referencing of sources is a vital safeguard against plagiarism.

3.8 Referencing

References must be consistent throughout the report. A complete reference list of all literature and other relevant documents must be given and properly referenced. All work cited in the main text should appear, fully referenced, in the reference list; all documents used should be cited in the main text.

By default, we recommend using the Harvard System for referencing, see:

<http://subjects.library.manchester.ac.uk/referencing/referencing-harvard>

However, there is variation in citation and referencing practice by discipline and field and you and/or your supervisor may have preferences. The best approach is to agree with

your supervisor the referencing system you will use at the first supervision and then stick to that throughout the process.

3.9 Appendices

Appendices, if necessary and if relevant, can also be included in your report. However, the use of appendices must be justified and legitimate: they should not just be a repository of disparate information which does not fit anywhere else; nor should they be used to avoid exceeding the word limit. Instead, appendices might usefully contain material such as, data sources consulted, outputs of exploratory analysis, tables summarising missing data. Code can also be placed in appendices as can descriptions of software packages where such cannot be easily referenced using a URL.

Bear in mind that appendix material is intended for reference only, and consultation is at the discretion of the reader. Critical information must therefore be included in the main text of the report. Appendices are aimed the reader who might want to dig a bit deeper – for example to look at the code underlying a particular model or data process but should never include materials that are necessary for understanding the main text. On the other hand, appendices should not be free floating. It is common to use footnotes at key points in the text to refer the interested reader on (e.g. “For more details of the missing data analyses see appendix C”).

3.10 Presentation conventions for text and formatting

There are several conventions to which you must adhere when submitting your completed report. These are listed below.

Title The title must be short, unambiguous and accurate.

Title page This should provide a statement as follows:

- Title of the project
- the following text: ‘A extended research project report submitted to the University of Manchester for the degree of xxx in the Faculty of Humanities’
- the year of submission (not including the month).
- the candidate’s student ID number; and
- the name of the candidate’s School (‘School of Social Sciences’).

Table of Contents A list of contents, giving all relevant sub-divisions of the report and a page number for each item (in Arabic numerals throughout). The final word count, including footnotes and endnotes, must be inserted at the bottom of the contents page. If illustrative materials are integrated within the text a separate list of illustrations should be prepared.

List of Illustrations The term ‘illustration’ refers to all tables, maps, plans, graphs, diagrams, photographs. The list of illustrations should provide number, title, and

page references. This usually appears on a separate page unless included in the table of contents.

Abstract All reports must include an abstract. This should usually be written when the report is otherwise complete. The abstract should precede the introduction so that the reader/examiner can quickly see what the text is about prior to more detailed reading. Typically, the abstract defines the problems the writer sets out to solve, the main procedures adopted, and the principal results and conclusions; it should occupy a single A4 page and can be single-spaced.

Acknowledgements Assistance given to the student in the preparation of their work must be acknowledged and could include the supervisor and any key individuals (other academics, individuals from the agencies under study etc) who have helped. Acknowledgments should not normally exceed one or two paragraphs.

Declaration A declaration stating that: 'No portion of the work referred to in this extended research project report has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning'.

Intellectual Property Statement All **four** of the following notes on copyright and the ownership of intellectual property rights must be included as written below:

- i. The author of this extended research project report (including any appendices and/or schedules to this report) owns certain copyright or related rights in it (the "Copyright") and s/he has given The University of Manchester certain rights to use such Copyright, including for administrative purposes.
- ii. Copies of this report, either in full or in extracts and whether in hard or electronic copy, may be made **only** in accordance with the Copyright, Designs and Patents Act 1988 (as amended) and regulations issued under it or, where appropriate, in accordance with licensing agreements which the University has entered into. This page must form part of any such copies made.
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take place is available in the University IP Policy (see <https://documents.manchester.ac.uk/display.aspx?DocID=24420>), in any relevant dissertation restriction declarations deposited in the University Library, The University Library's regulations (see <https://www.library.manchester.ac.uk/about/regulations/>) and in The University's Guidance for the Presentation of dissertations.

<i>Text</i>	Text should be formatted as double or 1.5 spacing, with a minimum font size of 12 for the main text. Single-spacing should be used for indented quotations of more than three lines, footnotes and references. Pages should be single-sided. Chapter headings, section headings should be bold and capitalised; sub-section headings should be bold.
<i>Page sizes</i>	The required size is A4 (297mm x 210mm).
<i>Page numbering</i>	Page numbering must consist of one single sequence of Arabic numerals (i.e., 1, 2, 3 ...) throughout the report. Page numbers must be displayed on all pages except the title page. The pagination sequence will include not only the text of the report but also the preliminary pages, diagrams, tables, figures, illustrations, appendices, references etc. Roman numerals must not be used for page numbering.
<i>Maps</i>	Maps should be clearly presented, with sources referenced. See Section 3.5 Graphical Material
<i>Diagrams and tables</i>	These should be clearly presented, properly sourced (unless self-generated), and explained in the text. See Section 3.5 Graphical Material
<i>Photographs</i>	See Section 3.5 Graphical Material
<i>Quotations</i>	Direct prose quotation exceeding three lines of text should be set out in a separate inset paragraph in single line spacing (indented about 25mm to the right and left of the main text), without inverted commas. Shorter quotations should be enclosed within the main text, in double inverted commas. If there are gaps in the quotation use three dots '...' to indicate where the words are left out. For all quotes, the author and page number must be stated. If it is desired to draw attention to a phrase in a quotation do this by italics but note in the reference whether any italics are in the original or have been added by you as author (e.g., Smith,

2012, emphasis in original; or Jones, 2012, emphasis added).

Footnotes

These should be used very sparingly, if at all. Where footnotes are deemed necessary, they should be of direct relevance to the topic. They should be placed at the bottom of the page. They should be numbered consecutively throughout the report. The font should be 9 point (in Times New Roman or similar). Text should be single spaced.

References

See section 3.8 of this document.

4. Assessment criteria: report

Criteria	Above 80% High distinction	70 – 80% Distinction	60 – 69 % Merit	50 – 59% Pass	Below 50% FAIL
Intellectual content and originality (25%)	High intellectual content, novel ideas and integrated excellently with the existing literature. It is likely to be publishable.	Very good intellectual content, some novel ideas, integrated well with the relevant literature. Possibly publishable	Good intellectual content, developed with reference to the literature.	Moderate intellectual content, with some integration with the literature.	Some evidence of intellectual input, limited integration with the literature.
Coherence of overall report (15%)	Excellent. Clear and logical progression through and between sections. All aims and outcomes of the project are very clear. Extensive literature -fully and correctly cited – used to provide balanced and informed view. Interpretation of literature provides a basis for project objectives.	Very good. Logical progression through and between sections. All aims and outcomes are clear. Literature clearly links to project objectives. Complete and correctly cited.	Good. Mostly logical progression through and between sections. Main aims and outcomes of the project are clear. Some link between literature and project objectives. Mostly complete and correctly cited.	Moderate. Progression through and between sections is uneven or unclear at times. Main aims and outcomes of the project are moderately clear. Little interpretation of literature and link to project objectives. Moderately complete and cited.	Poor. Little logical progression through and between each section. Some sections are not appropriate to the project as carried out. The main aims and outcomes of the project lack clarity. Incomplete or incorrectly cited. Difficulty in interpreting literature and using it as basis for project objectives.
Project design and methods (25%)	Excellent. Design and method totally aligned with objectives. Treatment of the data sophisticated.	Very good. Design and method aligned well with objectives. Treatment of the data appropriate.	Good. Any faults are minor and do not detract from the overall quality of the project. Treatment of the data appropriate.	Moderate. Minor faults which detract from the overall quality of the research, but most of the methods used are sound. Treatment of the data is mostly appropriate.	Poor. Some major faults which detract from the overall quality of the project. Methods used are partially appropriate or correct. Treatment of the data contains errors.
Results and analysis (25%)	Excellently presented. Results have been analysed and interpreted at a level suitable for publication.	Presented to a high standard, with no major flaws. With minor changes results and analysis are suitable for publication.	Well presented, with occasional flaws and minor errors only. Analysis & interpretation mostly sound.	Moderately presented, but with some major flaws or several minor errors. Analysis & Interpretation moderate.	Poorly presented, several major flaws and/or many minor errors. Analysis & interpretation contain significant deficiencies.
Overall presentation (10%)	Excellent throughout. All figures and tables clear with suitable legends and/or captions.	Very good throughout, with only minor shortcomings.	Good throughout, with no major flaws but occasional minor errors. Some figures / tables unclear.	A few major flaws and/or several minor errors. Several figures or tables of poor quality.	Some major flaws and/or frequent minor errors. Many poor-quality figures/tables.

Appendix 1 – MSc Data Science Extended research project Timetable

December and January	Note that the instructions in <u>this box</u> do not apply to students on the CSDI pathway. Your pathway lead will be in contact with you separately about your arrangements. - A set of extended research project outlines to be circulated to all students by the 15th December 2026. - Students who wish to develop their own project topic should submit a project outline to their pathway lead by 4PM on the 20th January 2026. - All students should submit a project preference form via this link https://airtable.com/apprUpJndD62tley1/shrlnW3W2b6jNsmJr - The form should be submitted by 4PM on the 31st January 2026. If you do not submit a form, we will assume that you have no preference as to projects.
February-June	Five meetings with supervisor, some of these may be group supervisions.
February	Allocation of projects – by 12 th February 2026.
March	The programme ethics form should be completed by 21st March 2026. See section 2.7 for more information. Project plans should be submitted via Turnitin on 31st March 2026.
April	If needed, an ethics decision meeting will take place in April with the programme director. If you are required to complete ethics review manager: https://submission-ethicalreview.manchester.ac.uk/ActivityForm/Index then that needs to be done by 2nd May 2026. Completion of that tool will automatically notify the relevant team.
July-August	Extra supervisory meetings may be arranged during the summer months after consulting your supervisor for their availability. Complete a Notice of Submission Form by 29th August 2026
September	You should submit an electronic copy of your project report in the approved format by 16:00 on Monday 7th September 2026. Reports must be submitted via Blackboard and not to your supervisor. Details of the submission process will be communicated by the Soss Assessment team in the final weeks before submission is due. See section 1.3 Submission arrangements and section 3. On Submission format for more information. You should submit your video by 16:00 on Monday 14th September 2026.

Appendix 2 –Preparing your project report

A2.1 Possible formats for the report

There are two possible approaches to structuring your research report. Which you chose will partly depend on the nature of your research and you should seek guidance from your supervisor about the best approach for you.

Format 1: An academic paper

1. Introduction
2. Theory/lit review
3. Methodology
4. Results
5. Discussion/Analysis

Format 2: A technical report

1. Introduction
2. State of the Art
3. Study Design
4. Results/Evaluation
5. Discussion

A2.2 Academic paper format.

The advice here is indicative rather than prescriptive it is worth emphasising that the precise content and chapter sequencing may vary, depending upon the nature of your topic. As with all other matters discuss this with your supervisor. You will be provided with a set of exemplar high quality extended research projects to review, and this should give you some idea about the range of possibilities.

Note that is generally not advisable to call sections generic things like ‘literature review’ or ‘results’; try to select section titles that give a clearer flavour of the focus of your research.

The Introduction

It is easy to underestimate the importance of the introduction; it is often left to the end, but they may not be wise for two reasons. Firstly, this is the first thing that your examiner will read and will set the tone for how they view the report. Therefore, putting some effort into producing high quality introduction will likely reap disproportionate value; this will not happen if you leave writing it to the last minute. Secondly, writing

this section early will help focus your mind on the material required for the final report, and in planning your time.

Your goal should be that the reader knows by the end of the introduction what they are about to read and why this piece of research is worth doing. A standard introduction usually covers three topics background, scope, and structure.

Background

The report should begin by outlining the background to your topic. This includes the description & understanding of topic/problems, awareness, and solutions to technical/scientific challenges. It could also include both the broad policy context (where relevant), and details of other research studies which have looked at the area of study. You must remember that this section should act as a general introduction to the study, and should therefore be short and snappy, avoiding too much detail.

Shape and scope of the study

The second section of the introduction should build upon the background context that you have outlined and explain in brief the purpose of the study and the broad questions and issues you will be addressing. These can be explained in greater detail at the beginning of the methods section, when, based on the earlier literature review, you state the formal aims and objectives of the study and outline any research questions to be addressed or hypotheses to be tested.

Structure

The third section of the introduction is normally an outline of the structure of the report. The precise structure adopted will be dependent on your pathway and the topic chosen and should be devised in consultation with your supervisor.

The theory section / Literature review

The second section is where you describe the context of your research and develop the narrative that explains your choice of research questions. This section will often be in the form of a literature review, although again the structure will vary according to the precise topic selected and should be discussed at length with your supervisor.

A literature review is something which confuses many students. In essence, the aims of a literature review are two-fold. First, it should bring the reader up to date on **previous research findings** in the field, with reference to your chosen topic. Second, it should pull out the key ideas and findings from past research and 'locate' your study within that broader body of knowledge.

Problematic or weak literature reviews tend to be ones which are:

- *disconnected from other aspects of the study.* This can occur when they are written early in the process and not revisited later. Ensure that the section leads logically to your selection of aims and objectives and the key themes are

referred to in your later primary research and analysis sections.

- *overly descriptive*. It can be tempting to spend a lot of time describing policy, but although this demonstrates awareness it does not enable you to pick up too many marks. Ensure that you spend time analysing the effects and developing this critique as you go.
- *lack a coherent argument*. Although you are reviewing what may be a broad span of literature you should still spend time trying to make it work as ONE narrative – consider the best order to introduce topics to enable the argument to logically develop, with each section building upon what has gone before.
- *lack a powerful end statement*. The literature review is important for scoping the study, but should also be designed to deliver an argument to justify YOUR extended research project – what is the research gap you are intending to address? And why is it important? This also helps you link to the following methodology section where you explain your selected research approach.

It is also vital that you avoid plagiarism, whether unintentional or deliberate. If you use ideas or quote a short passage from others' work – which is, of course, perfectly acceptable – you must acknowledge the source by full and proper referencing.

A useful starting point for literature reviews is to read a small number of core texts, and then trace back the more detailed articles cited.

Methodology

The methodology section might start with an outline the **conceptual framework** that you employ in the project. If this is the case, then you need to think of a conceptual framework in the following ways:

- It is a set of coherent ideas or concepts organised in a way that makes them easy to communicate to others.
- An organised way of thinking about the methods to be employed and the overall approach.

In the methodology section you would normally state the overall **aim of the project** followed by a set of **objectives, research questions and/or hypotheses**. The exact mix of these will depend on the nature of your study.

- **Aim**: this sets out the overall purpose of the study. It is a broad statement which explains what you are trying to achieve to a non-expert reader who may not be familiar with your topic area.
- **Objectives**: these are the specific operational targets which will assist in meeting the broad aims of the study. Since these objectives are clearly set out, they will be used to judge what you have been able to achieve at the end of your project. It is thus unwise to be over-ambitious by setting objectives which

are not realistically achievable. At the same time, devising clear objectives at the outset of your research will ensure that the work is sufficiently focused, and avoid the work being too generalised. Throughout your programme of research, you should constantly refer to your objectives to ensure that what you are doing, or writing is relevant, and to ensure that it will help to meet those objectives.

- **Research questions** are specific enquiries e.g., which factors within the Cancer60 data set are predictors of whether men in the UK over the age of 60 develop prostate cancer.
- **Hypotheses are specific propositions that are** to be tested, e.g., ‘urban deprivation in Manchester neighbourhoods between 2001 and 2015 was not affected by the level of spend on urban regeneration projects’. The veracity of this hypothesis might be tested, using models of data of local and national government spending linked to Index of Multiple Deprivation data for 2001, 2007, 2010 and 2015.

While it is important to clearly articulate the purpose of your research; take care not to devise an opaque and overly elaborate array of aims, objectives, research questions and hypotheses. Keep things simple and transparent.

Having generated relevant research questions and/or hypotheses, you must explain *how* you will go about answering or testing these. In other words, you must give details of the research methods to be used, outlining the overall **research design**, and specifying methods of data collection (e.g., sources of published data that have been used). You must **justify** the choice of your particular methodology, considering the chosen research topic: you should demonstrate why the methods selected are appropriate to answering a particular question, or investigating a given hypothesis.

Data

An important part of the methodology will be a description of the data that you have used for your study. The exact contents of this section will depend on the role that data plays in the project. Some of the questions that might be covered are:

- The source of the data
- The data generation processes. How is the data generated and how does that process affect the meaning of the data and its quality?
- Is the data a sample or complete representation of a particular population? What population is the data drawn from? If a sample what is the relationship between the sample and the population? How might that affect inferences?
- What pre-processing you did prepare the data for analysis wished to conduct? For example, did you clean the data and if so how. How did you take account of missing data?
- Were the data created through linking/matching multiple datasets together? If so, what linking methods did you employ?

- What features/attributes/variables are contained in the data?
- You might report descriptive analysis here – for example univariate tables/statistics – to give the reader an idea of what the data look like. Note however that if there are more than a small number of these then they should probably be placed in appendices.

Results and analysis/evaluation/discussion

In some cases, you might opt to have two sections devoted to 'results': a first one describing your main findings and outlining the results; and a second discussing the broader implications. Whether you have one or two sections is likely to depend upon the topic. Again, you will need to discuss this with your supervisor.

Whatever you decide, this/these sections form the core of your report. Having set the stage with your theory and literature review and carefully outlined your research process in the methodology section, you now reveal the results of the research itself and analyse and discuss those results.

The main point to emphasise here is that you need to *analyse*. Read back through each paragraph of your work: are you simply describing results or spending time expanding on the implications for theory, policy, or practice? It can be helpful to read back over your literature review at this stage to give you some key themes to add to your analysis, and to help increase the overall coherence of your work.

The details of your approach will again depend on your topic and pathway. In writing a suitably analytical and interpretative piece of work, it is important that you refer to your initial objectives and avoid being side-tracked by irrelevant detail or bogged down by the superfluous minutiae which surround any topic. At the same time, you must also take care to ensure that the information collected is not simply 'analysed' for its own sake, without identifying the implications for the study. Throughout this stage of the work, you should ask yourself what is the implication of a given finding for your research questions or hypotheses.

Conclusions

The concluding section or section should not merely summarise material already covered in previous sections. Instead, you must attempt to draw together the various messages to emerge from your review of the literature, and from your 'analysis' section(s). Again, it is important not to be descriptive, and to concentrate on the research questions posed at the outset of your study. You should try to highlight the implications of your study for both (a) research and knowledge of a particular topic area, and (where relevant) (b) policy and/or practice in relation to the topic area and field. You may also want to propose objectives/hypotheses that future research in this area could investigate/explore.

A2.3 Technical report format.

The advice here is indicative rather than prescriptive it is worth emphasising that the precise content and chapter sequencing may vary, depending upon the nature of your topic. As with all other matters discuss this with your supervisor. You will be provided with a set of exemplar high quality extended research projects to review, and this should give you some idea about the range of possibilities.

Note that is generally not advisable to call sections generic things like 'literature review' or 'results'; try to select section titles that give a clearer flavour of the focus of your research.

The Introduction

It is easy to underestimate the importance of the introduction; it is often left to the end, but they may not be wise for two reasons. Firstly, this is the first thing that your examiner will read and will set the tone for how they view the report. Therefore, putting some effort into producing high quality introduction will likely reap disproportionate value; this will not happen if you leave writing it to the last minute. Secondly, writing this section early will help focus your mind on the material required for the final report, and in planning your time.

Your goal should be that the reader knows by the end of the introduction what they are about to read and why this piece of research is worth doing. A standard introduction usually covers three topics background, scope, and structure.

Background

The report should begin by outlining the background to your topic. This includes the description & understanding of topic/problems, awareness, and solutions to technical/scientific challenges. It could also include both the broad policy context (where relevant), and details of other research studies which have looked at the area of study. You must remember that this section should act as a general introduction to the study, and should therefore be short and snappy, avoiding too much detail.

Shape and scope of the study

The second subsection of the introduction should build upon the background context that you have outlined and explain in brief the purpose of the study and the broad questions and issues you will be addressing. These can be explained in greater detail at the beginning of the study design section, when, based on the your evaluation of the state of the art, you state the formal aims and objectives of the study and outline question to be addressed or hypotheses to be tested.

Structure

The third subsection of the introduction is normally an outline of the structure of the report. The precise structure adopted will be dependent on your pathway and the topic chosen and should be devised in consultation with your supervisor.

The state of the art section

The presupposition of the technical report format is that you are in your project developing a new method, algorithm, or approach. So, your kicking off point is what are people in the field currently doing? i.e., what is the state of the art?.

This is something which confuses many students. In essence, the aims of this section are two-fold. First, it should bring the reader up to date on **previous research findings** in the field, that are relevant to the method or approach that you want to develop.. Second it should provide an indication of the standards achieved on critical metrics of existing approaches..

- methodology section where you explain your selected research approach.

It is also vital that you avoid plagiarism, whether unintentional or deliberate. If you use ideas or quote a short passage from others' work – which is, of course, perfectly acceptable – you must acknowledge the source by full and proper referencing.

Study Design

In the study design section you would normally state the overall **aim of the project** followed by a set of **objectives**. The exact mix of these will depend on the nature of your study.

- **Aim:** this sets out the overall purpose of the study. It is a broad statement which explains what you are trying to achieve to a non-expert reader who may not be familiar with your topic area.
- **Objectives:** these are the specific operational targets which will assist in meeting the broad aims of the study. Since these objectives are clearly set out, they will be used to judge what you have been able to achieve at the end of your project. It is thus unwise to be over-ambitious by setting objectives which are not realistically achievable. At the same time, devising clear objectives at the outset of your project will ensure that the work is sufficiently focused, and avoid the work being too generalised.

While it is important to clearly articulate the purpose of your research; take care not to devise an opaque and overly elaborate array of aims and objectives. Keep things simple and transparent.

Having generated your aims and objectives, you must explain *how* you will go about delivering these. In other words, you must give details of the to be used, outlining the overall **research design**, and importantly the evaluation methods that you will employ to assess the approach algorithm or method that your project has developed.

You must **justify** the choice of your particular design: you should demonstrate why the evaluation approach selected are appropriate to addressing the study objectives.

Data

An important part of the study design will be a description of the data that you have used for your project. The exact contents of this section will depend on the role that data plays in the project. Some of the questions that might be covered are:

- The source of the data
- The data generation processes. How is the data generated and how does that process affect the meaning of the data and its quality?
- Is the data a sample or complete representation of a particular population? What population is the data drawn from? If a sample what is the relationship between the sample and the population? How might that affect inferences?
- What pre-processing you did prepare the data for analysis wished to conduct? For example, did you clean the data and if so how. How did you take account of missing data?
- Were the data created through linking/matching multiple datasets together? If so, what linking methods did you employ?
- What features/attributes/variables are contained in the data?
- You might report descriptive analysis here – for example univariate tables/statistics – to give the reader an idea of what the data look like. Note however that if there are more than a small number of these then they should probably be placed in appendices.

Results and analysis/evaluation/discussion

In some cases, you might opt to have two sections devoted to 'results': a first one describing your main findings and outlining the results; and a second discussing the broader implications. Whether you have one or two sections is likely to depend upon the topic. Again, you will need to discuss this with your supervisor.

Whatever you decide, this/these sections form the core of your report. Having set the stage with your state of the art section and carefully outlined your research process in the study design section, you now reveal the results of the research itself and analyse and discuss those results.

Conclusions

The concluding section or section should not merely summarise material already covered in previous sections. Instead, you must attempt to draw together the various messages to emerge from your study. Again, it is important not to be descriptive, and to concentrate on the objectives set at the outset of your study. You should try to highlight the implications of your study for both (a) research and knowledge of a particular topic area, and (where relevant) (b) policy and/or practice in relation to the

topic area and field. You may also want to propose objectives/hypotheses that future research in this area could investigate/explore.

A2.4 Report Submission Check List

You need to check your draft for what might be termed continuity errors. Roughly speaking, this means checking that the whole text is consistent with itself from beginning to end. If you have changed some sections, there might be section headings to re-number, for example. To help you eliminate such errors, here is a checklist:

- are the headings and sub-headings in the contents list the same as those in the text?
- have you given lists of tables and figures as well as sections in your contents?
- are all sections, sections, sub-sections numbered consecutively? Numberings from earlier versions may persist and you may have two section 4s or no section 6. Using the automated Table of Contents wizard in Word can help enormously in this respect.
- are all the cross-references to other sections of the report correct?
- do all the references in the text have a corresponding entry in the reference list, with the same date as the reference in the text?
- where you refer to an article within an edited collection, have you included the full book reference, with editors, as well as the section reference?
- are all the references complete, i.e., have you included the publication date and place, as well as the publisher's name?
- tables: check that their numbers and titles are correct, and that references to them in the text are correct.
- figures: as for tables
- have you checked all the calculations in your tables? Have you got correct totals in the tables?
- is there enough labelling information in your tables and graphs? (e.g., if you refer to percentages, is it clear exactly what they are percentages of? Do you make it clear whether raw scores or percentages are being referred to?)
- have you calculated all the figures to the same number of decimal places?
- Have you been consistent in your use of abbreviations?

Appendix 3 Guidance Project Plan Guidance

A3.1 Intro

You should submit your project plan using the form provided. Plans should be submitted by the 1st April, 17:00 on Blackboard.

Your submission file should be named using the following format:

<pathway>-<student number>-project-plan.docx

- BM = Business and Management
- CSDI = Computer Science Data Informatics
- EAN = Environmental Analytics
- MAT = Maths
- SAN = Social Analytics
- UA= Urban analytics

So, if your pathway were **Maths** and your student number were **12956267**, your file name would be:

MAT-12956267-project-plan.docx

The plan should cover:

- Non-technical project summary
 - A summary in plain English of the proposed project.
- Research questions
 - The Aim of the project and up to four Research questions or Objectives
 - A simple description of the projects aim, and up to 4 research questions or objectives. These should be clear and concise.
- Methodology
 - Description of Methodology to be used (up to 400 words)
 - How will the project goals be achieved? What work will be done (or at least provide a starting point for the work)? Include a description of the data.
- Timetable
 - Up to 250 words (and a diagram which can be added as a separate page).
 - Use a Gantt chart or similar visualisation (on an extra page) in combination with the text to describe the project timeline. Deadlines and milestones should be identified.
- Risk assessment and management
 - Describe any risks to the project succeeding and how you will manage them (150 words).
 - Identify potential risks to the project, whether technical or something else, and indicate what steps you will take to avoid/mitigate those risks.

These will be marked by your pathway lead, and you will be given feedback indicating where the plan can be improved. You can receive input from your supervisor before submission. If you have any queries or concerns, then I have office hours available every day accessible via:

<https://me-office-hours.youcanbook.me/>

A3.2 The Rubric

Non-technical project summary (15%)

A summary in plain English of the proposed project.

Unacceptable: 0 - 40%

It is unclear what the project is about. The written quality is not acceptable at master's level.

Problematic: 40 - 50%

The subject area of the project is presented but in an unconvincing manner. An attempt to describe it has been made although this might be less than satisfactory. Perhaps technical terms have been used without explanation. Aspects of the writing stand in the way of comprehension.

Satisfactory: 50 - 60%

The subject area of the project is reasonably clear with a purpose that basically make sense. The written quality is understandable although the writing contains errors.

Good: 60 - 70%

The subject area is well described with purpose clearly stated. The written quality is mostly appropriate, but possibly some minor language, style or English usage issues.

Very Good: 70 - 80%

The subject area is described well with clear non-technical English. The purpose of the project is clearly stated and coherent. The written quality will be very good with only the very occasional minor errors.

Excellent: 80 -100%

The subject area of the project will be compellingly presented. The purpose of the project is completely clear and entirely self-consistent. High quality writing with an appropriate structure and style, clear sentences with correct English non-technical usage and punctuation.

Research questions (10%)

A simple description of the projects aim and up to 4 research questions or objectives. These should be clear and concise

Unacceptable: 0 - 40%

No research questions or aims and objectives have been provided, or they are not understandable in terms of the project.

Problematic: 40 - 50%

An attempt to define the research questions or aims and objectives will have been made although these might be less than satisfactory.

Satisfactory: 50 - 60%

Research questions or aims and objectives have been described basically make sense but are not a coherent set and/or do not have a clear relationship with the project topic.

Good: 60 - 70%

Aims and objectives clearly stated but are perhaps not a completely coherent set.

Very Good: 70 - 80%

Aims and objectives are clearly stated and coherent

Excellent: 80 -100%

Aims and objectives are completely clear and entirely self-consistent.

Description of Methodology to be used (40%)

How will the project goals be achieved? What work will be done or at least provide a starting point for the work? Include a description of the data.

Unacceptable: 0 - 40%

There is only a very minimal, if any, attempt to describe the data and methods that will be employed to undertake the work of the project. The written quality is unlikely to be acceptable at master's level.

Problematic: 40 - 50%

Some attempt has been made to describe the methodology to be used in the project. However, the description is shallow, and it is not clear whether it has been fully understood. Aspects of the writing may have stood in the way of comprehension.

Satisfactory: 50 - 60%

A reasonable attempt to describe the research methodology is provided. It is mostly clear, but there may be some language or style issues; the main technical points appear to be understood.

Good: 60 - 70%

The research methodology to be adopted is clear as is the starting point for the research work. The written quality is mostly appropriate, but possibly some minor language, style or English usage issues.

Very Good: 70 - 80%

The research methodology is both clear and demonstrates a very good mastery of the techniques to be employed. It is well written with very few mistakes in written English, technical terms and acronyms are always defined.

Excellent: 80 -100%

The methodology to be used during the project is described in real depth. It is clear that the student has a deep understanding of the techniques to be used. High quality writing with an appropriate structure and style, clear sentences with correct English usage and punctuation.

Timetable (20%)

Use a Gantt chart or similar visualisation (on an extra page) in combination with the text to describe the project timeline. Deadlines and milestones should be identified.

Unacceptable: 0 - 40%

A project plan, if present, is either unreadable or make no sense. No milestones have been identified.

Problematic: 40 - 50%

A timeline is provided but contains errors, for example too many/too few sub-tasks or an inappropriate time discretisation. The timeline does not appear thought through. Milestones may not be clear.

Satisfactory: 50 - 60%

A reasonable attempt to produce a time plan has been made. Some milestones have been included and the sub-tasks / time discretisation appear plausible. The timeline makes reasonable sense.

Good: 60 - 70%

A good attempt at a time plan has been made. Some milestones (~3-5) have been identified and the sub-tasks / time discretisation appears appropriate. The timeline appears to match the objectives of the project.

Very Good: 70 - 80%

A clear project plan is presented. The timeline is entirely reasonable for the work of the project matching the aims and objectives. Milestones are unambiguously identified and appropriate in content and number (typically 3-5). The written part will be of a high standard.

Excellent: 80 -100%

A comprehensive and thoroughly readable plan will have been presented with an appropriate number of individual activities and timeline discretisation. Milestones are clearly identified. There is a description of the plan in the text; the written quality is of a high standard.

Risk assessment and management (15%)

Identify potential risks to the project, whether technical or something else, and indicate what steps you will take to avoid/mitigate those risks.

Unacceptable: 0 - 40%

At most there is a one or two sentence considering risk. There is no attempt to justify the statements made or conclusions reached regarding risk.

Problematic: 40 - 50%

There is no real evidence that risk has been appropriately considered in the context of the project. The written quality might be poor making it difficult to understand the work.

Satisfactory: 50 - 60%

A reasonable attempt, probably a few sentences, has been made to consider risk factors with at least some relevance to the project. Some basic mitigation measures may be described. The written quality may contain errors, but these will not be at a level that impedes understanding.

Good: 60 - 70%

A decent attempt has been made to understand potential risks which might affect the work of the project. There will be some measures described to manage risk. The written quality will be satisfactory and understandable but not necessarily flawless.

Very Good: 70 - 80%

Risk considerations have been taken seriously and provides evidence that an analysis of risk has been undertaken. Measures to manage risk are present. The written quality will be consistent with the minimum required for a distinction level MSc dissertation.

Excellent: 80 -100%

A thorough and comprehensive description of risk entirely appropriate for the work of the project. Key risk factors have been considered and possible actions to avoid/mitigate risk identified. The written quality will be of a high standard.

Appendix 4: Guidance on the Additional Materials

The companion piece of assessment for your extended research project report is the additional materials. These should be designed to provide and demonstrate reproducibility for the research that you are reporting on.

A4.1 Content

We have avoided being overly prescriptive about this – what is needed will be project dependent.

A litmus test would be something like: “could a competent researcher who does not have access to my report use these materials to exactly reproduce what I have done”. If your answer to this is yes, then you have a complete set of additional materials. Just as a Figure or a Table within any report should be comprehensible on its own without reference to the original text, so the additional materials should allow reproduction of your research without reference to the report itself. This means that you may wish to include a copy of your reported analytical output in the additional materials – these would allow somebody attempting to reproduce your work to verify whether they have done so.

Content might include:

- a) A technical appendix describing, in sufficient detail, the procedures used to run all the analysis in the ERP report. These will include all process you used to capture/collect the data, all preprocessing and EDA that you have reported, models that you have run etc. Note you do not and should not report on analysis that you have run as part of your own process, but which don’t appear in the report. The goal is reproducibility not explication of your process.
- b) A (public) GitHub repository of code and any materials supporting the running of that code. If the amount of code is small, then this could instead be added directly to the technical appendix.
- c) Remember to include clear information of any packages that you used, parameter settings etc.

Note: to avoid data protection complexities, you can assume that the researcher does already have access to the source data that you used in your research – i.e. you do not need to make the data themselves available. You may want to include a link to the data in the appendix if they are public or to the source/means of access if they are not.

A4.2 Length

There is no length limit for the additional materials. However, do not take this as license to include anything and everything that might be relevant or “everything that you did”. Excessive material volume can obfuscate reproducibility rather than enable it. Think

carefully about what material is needed by somebody wanting to reproduce your work and include just that and no more. Quality of the material rather than the quantity of it is critical. Clarity of any instructions, good and proportionate use of annotation, appropriate use of flow diagrams, tables and other means of structuring the information are important.

A4.3 Assessment Rubric for additional materials

Levels	Reproducibility 75%	Clarity of Presentation 25%
High Distinction	The materials are precisely sufficient to allow reproduction of the research project with no superfluity	The materials have been produced in attractive and accessible form. Any text is coherent and well written. Any figures tables and visualisations clear with suitable legends and/or captions.
Distinction	The materials are sufficient to allow reproduction of the research project, with a little superfluous material	The materials have been produced in attractive and accessible form. Any text is coherent. Figures, tables and other visualisations are mostly clear with suitable legends and/or captions.
Merit	The materials are generally sufficient to allow reproduction of the research project, with some superfluous material and possibly some small gaps in the instructions.	The materials have been produced in a generally accessible form. Good throughout, with no major flaws but perhaps occasional lack of clarity. Perhaps some figures / tables are unclear.
Pass	The materials contain some deficiencies meaning they are not sufficient to allow reproduction of the research project, with significant amount of superfluous material and some gaps in the instructions.	A few major flaws and/ or several minor places where there is a lack of clarity. Several figures or tables of poor quality. Perhaps insufficient use of visualisations to support comprehensibility.
Fail	The materials are insufficient for reproducibility.	The materials are poorly presented. Some major flaws and/or frequent lack of clarity. Poor use of figures/tables and other supporting visualisations.

Appendix 5: Extended research project: Guidance on Video Assessment

The final assessment element for your extended research project is the video. It's a short (no more than about five minutes) recorded presentation of your work.

This is an alternative form of disseminating the outcomes of your project. The basis of this complement is that you will have to present the results of project to non-cognate audiences. Perhaps the management of the company, a client or a user.

How you structure your video is entirely up to you and will depend very strongly on the subject matter of your project. But there are three likely formats:

- Screencast – here the video part will be capture of your screen, this likely to be the right approach if you want to demonstrate some software. Perhaps here the imagined audience is a user.
- Video poster – a format where you have a single slide which contains a summary of the project, and you talk over possible travelling round the poster as you speak. This will lend itself to presentation to clients.
- Traditional slides and talking head – perhaps the imagined audience is your company's management board.

A5.1 Structure

Think carefully about what you can present that will effectively communicate what your project is about and what you have achieved.

Like every other submission you make, ensure that you introduce yourself and the project before moving to the body of the presentation.

The presentation should have an explanation of what you've done, you may not have time to include details of how you did it.

You should show results. It's important that you communicate clearly, so think carefully about the materials you want to present, the order you'll present them and how you'll present.

A5.2 Preparation

There are many ways of creating a video.

At the simple end of the spectrum, you could animate a PowerPoint presentation and record a commentary. This may be appropriate for some styles of project. At the other end you might use a multifeatured video camera and editing suite. For many people though, perfectly satisfactory results can be achieved using a mobile phone on a stand.

To create the video, you first need to write a script. You'll need to plan the story you want to tell and how to tell it; what information will you present and in what order. You may find it helpful to write out the words you're going to say.

Then assemble your materials, there might be animations you've made in PowerPoint, or simple sequences of slides, you might have video clips or screen captures.

Finally edit them into the video.

Some pitfalls to avoid.

- 1) Don't try to cram too much in either visually or verbally. Your goal is to communicate some key messages about your project to an audience who are (probably) not data scientists. Keep it simple.
- 2) If you are doing the "talking head with slides" format avoid the obvious mistake of placing yourself in front of material that you want the audience to see. If you have not got room on a slide for you to be standing in front without overlapping slide content then it might be that you have too much content on the slide!

When you have a complete video show it friends – do they understand from your

A5.3 Tools you can use

There are many free tools. The following are just a few suggestions.

Creating Videos

There are numerous ways in which you can create your video. Below are some of the more common options. These could be used individually or in combination (for example you could create video using their smartphone and then add this to an Adobe Express video).

Using Teams

A simple way to make a video recording or screencast is to record a Zoom or Teams meeting (with only you in the meeting). You can share and record your screen, as well as your own audio and video. NB Zoom can also be used in this way.

[Teams for UoM students](#)

[Recording a meeting](#)

[Downloading a recording](#)

Using PowerPoint

From within PowerPoint you can record a presentation, including narration and speaker video.

[PowerPoint for UoM Students](#)

[Recording in PowerPoint](#)

[Recording your screen in Powerpoint \(YouTube video\)](#)

Using Smartphones

You can use your smartphone to create video. You'll find some really useful tips about framing, lighting, sound and format on the [Faculty of Biology, Medicine and Health student support pages](#).

Using Adobe Express

Adobe Express is a freely available tool which includes options for video creation and editing. It is somewhat limited but can give professional looking results without too much technical expertise. You can create video from scratch using existing media (e.g. stock images and sound files) or use it to compile your own content (e.g. your own video and/or audio, images and text).

How to access

You should [sign up](#) with an email address, or using Facebook, Google or Apple.

Guides

[Introduction to Adobe Express for Students](#)
[Creating and editing videos in Adobe Express](#)

Using other free tools

Freely available tools like [ScreenPal](#) will allow students to create short screencasts. Alternatively, you could use these free Chrome browser extensions: [screencastify](#), [vimeo record](#).

- Screencapture Software
- Loom: www.loom.com
- OBS: obsproject.com/
- QuickTime

Video editing Software

Using ShotCut (free, open source video editor)

See the following video tutorials for a quick guide to editing in ShotCut:

- <https://shotcut.org/tutorials/>
- <https://shotcut.org/tutorials/minitutorials/>

A text based guide is available at <https://en.flossmanuals.net/shotcut/full/>

Using Adobe Express

See <https://new.express.adobe.com/webpage/2FPY3MJSEztNI>

Using other free tools

Freely available tools like [ScreenPal](#) will allow students to edit videos.

Other popular free video editing tools include: [Imovie](#), [Win10 Video editor](#), [OBS](#)

- iMovie on the Mac
- Openshot: www.openshot.org
- Lightworks: www.lwks.com
- Kdenlive <https://kdenlive.org/en/> (Linux, Windows, Mac)

A5.4 Notes on submission

Submission notes will be provided nearer the submission deadline.

A5.5 Notes on assessment

The video counts for 10 percent of the marks for the extended research project. It will be marked by the student's pathway lead (using the rubric below) and moderated by the programme director.

Assessment area	High Distinction	Distinction	Merit	Pass	Fail
AV quality 20%	The use of sound and visuals is outstanding and complements the content	The use of sound and visuals is excellent and appropriate for the content	The use of sounds and visuals is excellent is satisfactory for the content	Sound and/or visuals are adequate but do not complement the content	Sound and/or visuals are poor and do not complement the content
Communication 50%	The video communicates clearly what the project was about in a highly engaging manner	The video communicates clearly what the project was about in a highly engaging manner	The video communicates what the project was about but perhaps overemphasises some details	The video communicates some of the points of the project but is perhaps not sufficient for a full understanding	The video does not explain the project or its findings well
Delivery 30%	The delivery is of a strong with excellent pace and narration	The delivery is good with appropriate pace, well narrated and	The delivery is reasonable with satisfactory pace and narration and	The delivery is adequate but the pace being too fast or too slow and	The delivery is disjointed, and the pace is too fast or too

	and within limits	timed the	timed within the limits	timed within the specified limits	perhaps the video is too long or short.	slow and the video is too long or short.
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